

MANKATO, MN, USAWMO: **726585**Lat: **44.217N**Lon: **93.917W**Elev: **1021**StdP: **14.16**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **14954****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.1	-21.3	1.8	-10.2	-16.6	2.3	-7.2	31.0	13.7	28.0	12.2	9.0	310	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	89.9	73.7	86.4	72.1	83.6	70.8	77.1	85.5	74.9	82.8	73.0	80.8	13.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	133.8	83.0	72.3	124.3	80.7	70.3	115.8	78.6	41.3	85.2	39.0	82.9	37.2	81.0	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.8	24.2	20.9	-16.5	95.0	5.6	3.3	-20.6	97.4	-23.9	99.3	-27.0	101.2	-31.2	103.6		
(5)			-17.3	80.2	5.3	2.3	-21.1	81.8	-24.2	83.2	-27.1	84.5	-31.0	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	15.4	19.3	32.4	46.7	58.6	68.7	72.3	69.6	62.7	49.2	34.8	21.0		
	DBStd	22.33	13.20	12.76	12.56	9.93	8.86	6.64	5.42	5.32	8.29	9.40	11.17	12.09		
	HDD50	4190	1071	860	561	172	22	0	0	0	7	130	469	898		
	HDD65	7586	1536	1280	1012	551	239	37	7	17	140	497	907	1363		
	CDD50	2743	0	0	14	75	289	562	690	608	387	105	13	0		
	CDD65	663	0	0	1	4	40	149	232	159	71	7	0	0		
	CDH74	5362	0	0	4	57	436	1214	1937	1066	574	72	2	0		
	CDH80	1547	0	0	0	15	149	388	590	242	145	18	0	0		
	WSAvg	10.3	11.0	11.2	11.2	12.1	11.3	9.7	8.1	7.6	9.2	10.7	11.2	10.6		
	PrecAvg	33.9	1.0	0.9	2.0	3.0	4.0	5.1	4.3	4.7	3.3	2.6	1.8	1.2		
	PrecMax	43.9	2.8	2.1	3.9	5.8	7.3	8.4	7.4	9.8	10.0	6.1	4.6	3.8		
	PrecMin	23.0	0.1	0.1	0.3	0.7	1.5	3.0	1.7	1.6	0.7	0.6	0.2	0.2		
	PrecStd	4.9	0.6	0.5	0.9	1.4	1.7	1.7	1.8	1.8	2.1	1.6	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	53.1	70.2	81.4	90.9	92.9	92.6	90.0	89.7	82.1	68.5	53.2		
		MCWB	38.5	47.1	57.7	61.7	68.2	73.4	75.8	77.1	70.5	65.2	55.8	48.2		
	2%	DB	38.8	45.1	62.5	72.6	83.8	88.3	89.7	84.7	84.0	73.6	61.4	44.9		
		MCWB	35.0	40.5	53.9	56.3	64.1	71.4	76.5	74.4	70.0	60.7	52.0	40.2		
	5%	DB	36.1	39.4	55.2	67.5	79.0	84.1	86.1	82.1	81.0	69.6	55.1	38.9		
		MCWB	33.6	35.9	48.6	53.8	62.0	68.9	74.1	71.8	67.0	58.2	47.6	35.5		
	10%	DB	33.6	36.4	48.7	63.0	73.1	81.1	82.4	80.6	76.9	63.7	50.2	36.2		
		MCWB	31.4	33.5	43.6	50.8	58.7	66.9	71.5	70.1	65.8	54.0	44.1	33.5		
	0.4%	WB	38.8	47.3	61.0	63.3	71.8	76.8	81.0	79.5	75.4	68.3	58.3	49.3		
		MCDB	43.1	52.3	66.4	77.1	83.4	87.6	88.9	85.9	83.6	76.5	65.9	53.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.5	40.8	55.1	59.0	67.4	74.0	78.0	76.2	72.6	62.8	53.1	40.1		
		MCDB	38.2	44.9	61.2	69.5	79.1	83.9	86.7	83.0	80.5	71.0	59.9	44.1		
	5%	WB	33.4	36.7	49.8	55.4	64.3	71.4	75.6	73.9	69.9	59.0	48.4	36.1		
		MCDB	35.8	40.0	55.6	65.2	74.1	81.8	83.7	80.6	77.5	67.1	54.8	39.1		
	10%	WB	31.1	33.5	44.1	51.9	61.5	68.9	73.6	71.7	67.4	55.3	44.4	33.3		
		MCDB	32.8	36.2	49.6	61.1	71.6	78.2	81.2	78.3	74.4	63.0	49.9	35.8		
	5% DB	MDBR	14.6	15.6	15.4	18.3	18.5	17.5	17.4	17.0	19.4	17.8	15.2	13.5		
		MCDBR	15.7	17.0	23.4	27.3	26.3	22.0	19.5	18.6	23.4	25.7	22.8	17.8		
		MCWBR	13.2	13.8	16.0	15.7	12.6	10.6	11.1	10.9	12.1	14.5	15.1	14.5		
		MCWBR	15.2	15.9	22.2	23.4	21.4	19.2	17.9	17.1	19.8	21.8	21.5	16.7		
	5% WB	MCWBR	13.2	13.2	15.9	15.1	11.6	11.2	11.3	11.3	12.1	14.2	15.1	13.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.286	0.311	0.368	0.390	0.406	0.413	0.407	0.374	0.323	0.288	0.266		
	taud		2.449	2.382	2.434	2.320	2.319	2.319	2.298	2.299	2.361	2.506	2.520	2.460		
	Ebn at Noon		278	289	295	284	279	274	271	268	269	272	267	266		
	Edh at Noon		22	29	31	38	40	40	40	39	34	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		566	903	1181	1460	1707	1905	2033	1693	1341	872	566	416		
	RadStd		66	99	119	152	134	154	82	124	96	115	52	38		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page